Ahmedabad University

CSE 631: MachineLearning

Weekly Report(8)

(Group - 9) Learning Machine

Section 1

Submitted to faculty: Prof. Mehul Raval

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<u>Weekly Progress Report - 8</u>

Machine Learning–Based Automated Stone Counting System for Embroidered Fabrics

1. Things that have been accomplished this week.

This week was dedicated to the further development of the detection pipeline in terms of moving away from a CNN-based model to a more robust object detector (YOLO) and to enhancing the training stability through the application of new methods, such as OHEM, focal loss, and data augmentation.

1.1 Transition from CNN to YOLO

- We transferred to the YOLO (You Only Look Once) object detector model, based on the constraints we saw when using CNN + sliding window method (high computation and slower inference).
- YOLO enabled:
    - Object detection in real-time.
    - Sliding window-free end-to-end detection.
    - Localization with bounding boxes.
    - Simultaneous classification and detection
- This greatly enhanced speed and efficiency of the system.

1.2 YOLO Model Implementation

- Applied a YOLO-based detector:
    - Single forward pass processing of input image.
    - Detection of stones in a grid.
    - Confidence score prediction of bounding boxes.
    - Probability prediction on the class of each region detected.
- Advantages observed:
    - The inference is faster than the CNN sliding window.
    - Reduced redundancy in detection
    - Scalability to large images.

1.3 Image Preprocessing and Data Augmentation

- Improved preprocessing pipeline to better generalization:
- Preprocessing techniques:
    - CLAHE as a contrast enhancement.
    - Image normalization
    - Downsizing to YOLO input size.
- Data augmentation applied:
    - Rotation and flipping
    - Scaling and cropping
    - Intensity and change of contrast.
    - Noise addition
- Benefits:
    - Increased dataset diversity
    - Reduced overfitting
    - Enhanced resistance to changes in lighting and texture.

1.4 Focal Loss Implementation

- Added Focal Loss to manage the imbalance in classes and challenging samples.
- Key benefits:
    - Reduced impact of easy negative samples
    - Concentrated learning of stones that are hard to categorize.
    - Better performance on dense regions.
- Result:
    - Higher recall balance and precision.
    - Better identification of small stones and overlapping stones.

1.5 Online Hard Example Mining (OHEM)

- Used OHEM (Online Hard Example Mining) in training:
- Process:
    - Model determines hard samples (high loss regions).
    - Gives these samples a higher priority in training.
- Advantages:
    - Improved model robustness
    - Minimized false positives.
    - Improved management of complicated embroidery designs.

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Starting training for 20 epochs...
Focal Loss: alpha=0.75, gamma=2.0
OHEM ratio: 0.5 (keeping hardest 50% of negatives)
================================================================
Epoch   1/20 | Loss 0.0577/0.0207 | F1 0.232/0.346 | P 0.391 R 0.310 | LR 9.94e-05 | 4.3s
  ★ New best model saved (val F1=0.346)
Epoch   2/20 | Loss 0.0237/0.0205 | F1 0.587/0.431 | P 0.500 R 0.379 | LR 9.76e-05 | 4.4s
  ★ New best model saved (val F1=0.431)
Epoch   3/20 | Loss 0.0238/0.0154 | F1 0.592/0.506 | P 0.400 R 0.690 | LR 9.46e-05 | 4.3s
  ★ New best model saved (val F1=0.506)
Epoch   4/20 | Loss 0.0157/0.0150 | F1 0.651/0.563 | P 0.476 R 0.690 | LR 9.05e-05 | 4.2s
  ★ New best model saved (val F1=0.563)
Epoch   5/20 | Loss 0.0163/0.0164 | F1 0.648/0.610 | P 0.600 R 0.621 | LR 8.55e-05 | 4.3s
  ★ New best model saved (val F1=0.610)
Epoch   6/20 | Loss 0.0146/0.0151 | F1 0.697/0.590 | P 0.562 R 0.621 | LR 7.96e-05 | 4.3s
Epoch   7/20 | Loss 0.0125/0.0138 | F1 0.756/0.625 | P 0.571 R 0.690 | LR 7.30e-05 | 4.4s
  ★ New best model saved (val F1=0.625)
Epoch   8/20 | Loss 0.0119/0.0128 | F1 0.758/0.690 | P 0.690 R 0.690 | LR 6.58e-05 | 4.3s
  ★ New best model saved (val F1=0.690)
Epoch   9/20 | Loss 0.0133/0.0120 | F1 0.733/0.656 | P 0.625 R 0.690 | LR 5.82e-05 | 4.3s
Epoch  10/20 | Loss 0.0105/0.0109 | F1 0.713/0.677 | P 0.636 R 0.724 | LR 5.05e-05 | 4.2s
Epoch  11/20 | Loss 0.0136/0.0120 | F1 0.671/0.667 | P 0.645 R 0.690 | LR 4.28e-05 | 4.3s
Epoch  12/20 | Loss 0.0122/0.0115 | F1 0.732/0.667 | P 0.679 R 0.655 | LR 3.52e-05 | 4.3s
Epoch  13/20 | Loss 0.0118/0.0115 | F1 0.751/0.655 | P 0.655 R 0.655 | LR 2.80e-05 | 4.3s

Early stopping at epoch 13 (no improvement for 5 epochs)

Training complete. Best val F1=0.690 at epoch 8
```

1.6 Improved Detection Accuracy

- With YOLO + Focal Loss + OHEM:
  - Improved localization of stones.
  - Less overlapping and duplicate detections.
  - Better performance in solid stone areas.

1.7 Model Evaluation

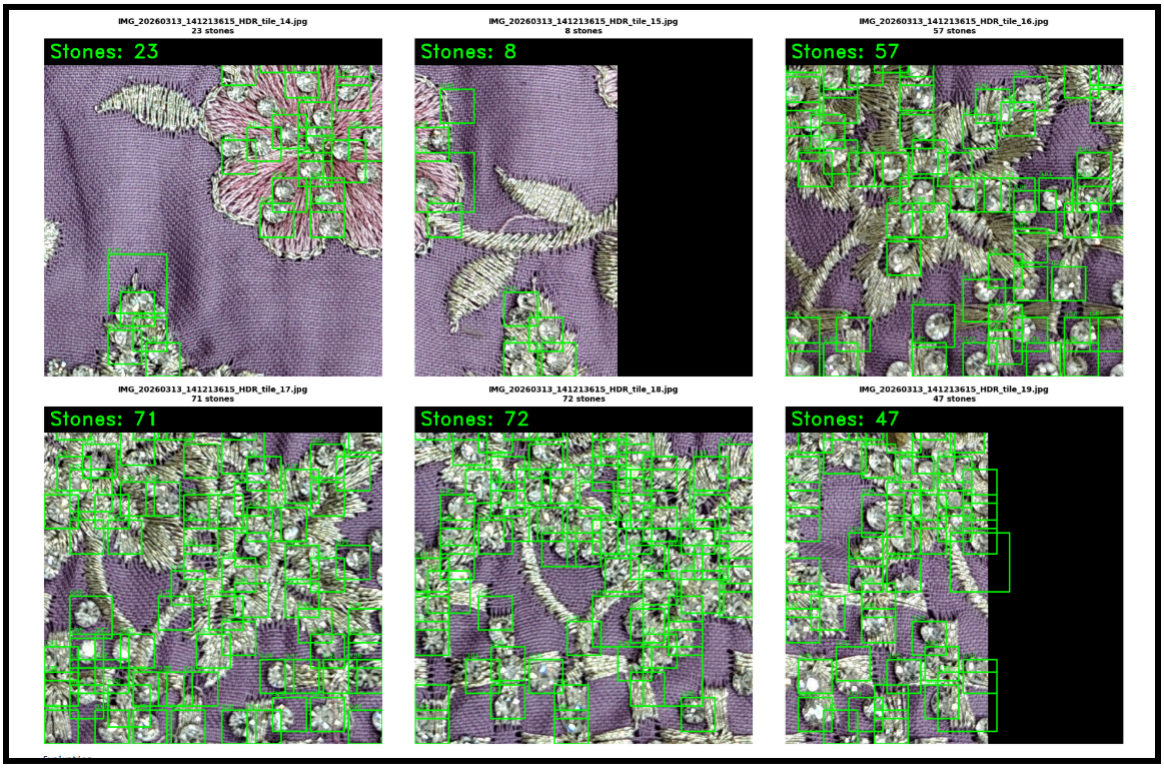
- Evaluated model using:
  - Accuracy
  - Precision
  - Recall
  - F1-score
- Observations:
  - YOLO was faster and more accurate than CNN-based pipeline.
  - Major increase in real-time detection performance.
  - Increased performance on high density and complicated embroidery images.

1.8 Key Findings

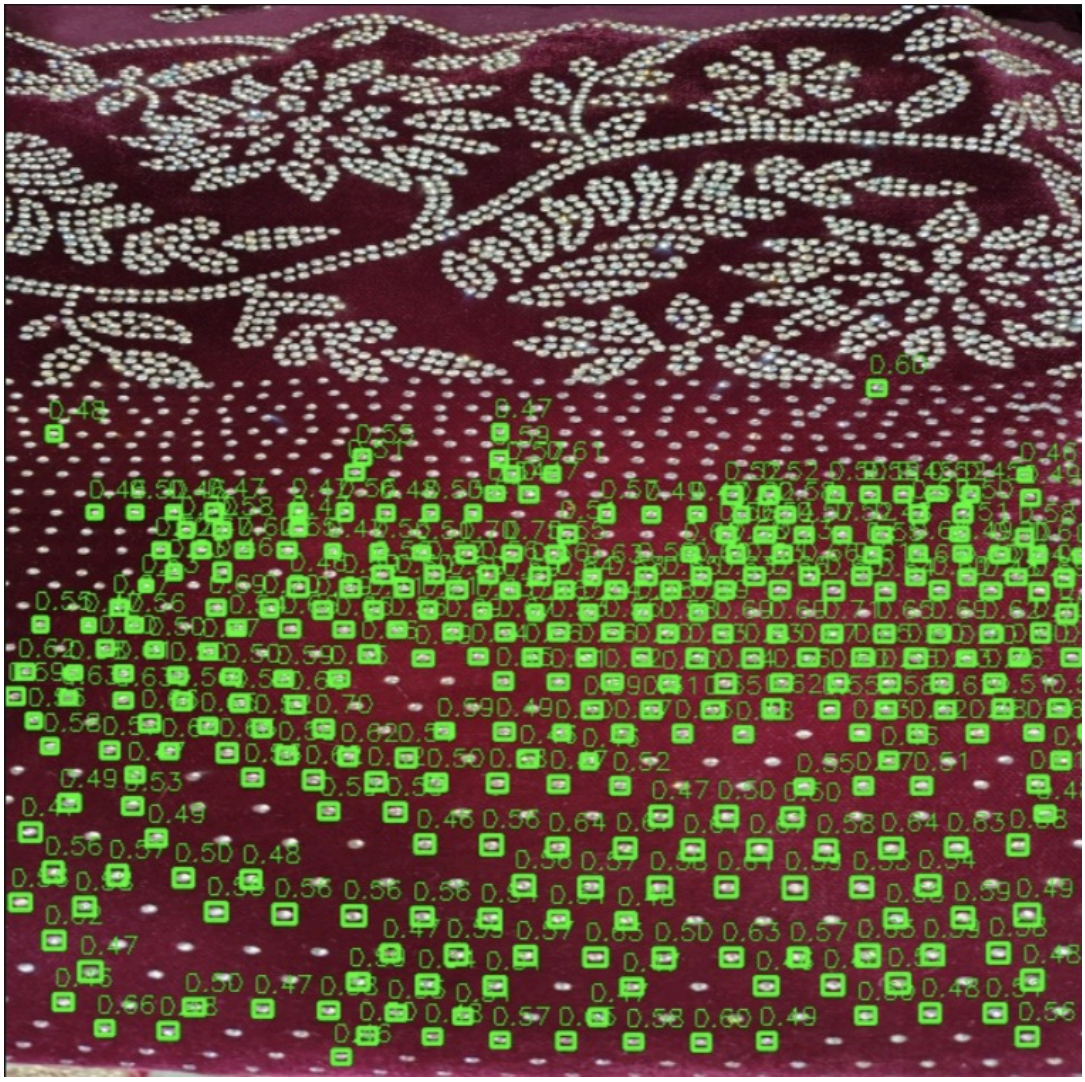
- YOLO will be the best option over CNN + sliding window in this problem.
- Generalization is aided by data augmentation.
- Focal Loss is efficient in dealing with class imbalance.
- OHEM enhances strength through emphasis on challenging samples.
- The joint method yields a huge improvement in the detection performance.

RESULTS:

Using CNN:



Using Yolo:



2. Activities to be done within the next week.

- Adjust the parameters of the YOLO model (learning rate, batch size, epochs)
- Try new versions of YOLO (YOLOv5/YOLOv8).
- Maximize detection speed to deploy in real time.
- Enhance the accuracy and localization of bounding boxes.
- Tune hyperparameters of focal loss and OHEM.
- Test model with more diverse and large datasets.
- Work on deployment pipeline (integration into application)
- Investigate methods of compressing models to run more quickly.